

Sunny Zhang

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Technical Skills: C++/C#/C, Python, ROS 1/2, linear control, adaptive control, computer vision, robotic simulation

EDUCATION

University of Toronto, Toronto, ON, CA

September 2022 – May 2027

Bachelor of Applied Science and Engineering, Engineering Science (Robotics Engineering) + PEY

Thesis: Adaptive Friction Model Controllers for Cable Driven Instruments for Transoral Robotics Surgery

Relevant courses: Adaptive Control and Reinforcement Learning, Controls Systems, Linear Control Theory, Robot Modelling and Control, Computer Vision for Robotics, Operating Systems, Digital and Computer Systems (FPGAs, FSMs, Verilog)

PROFESSIONAL EXPERIENCE

Vision Team Co-op Student, *Martinrea International Inc.*

Sept. 2025 – Sept 2026

Toronto, ON, CA

- Created, led, and owned a project developing vision guided industrial robot (FANUC) applications for automotive manufacturing
- Created a generalized controls library in C++** for industrial manipulators with communication over Ethernet/IP and Python bindings for computer vision guided control including **point cloud stitching and registration**
- Developed vision guided pick and place algorithms for industrial cobot using point cloud registration guidance
- Developed a custom PCB generalizing communications from raw PLC input/output to USB protocol to support end-of-life manufacturing system integration with computer vision systems, significantly reducing adoption costs

Robotics Intern, *Moon Surgical*

May 2025 – Sept. 2025

San Carlos, CA, USA

- Built robotic simulation features in C++** key to pre-release testing of features for the Maestro robotic surgery platform
- Conducted research and development of cutting-edge **force detection and disturbance rejection algorithms** to improve surgeon interaction with the Maestro

Student Researcher, *Center for Image-Guided Innovation and Therapeutic Intervention (CIGITI)*

Sept. 2023 – Sept. 2025

Toronto, ON, CA

- Developing cutting edge **friction modelling and adaptive control algorithms** for a cable driven surgical instrument
- Led the research and development of a novel mm-scale elbowed and wristed surgical tool and preliminary prototypes of a novel surgical system specialized for minimally invasive trans-oral robotic surgery, accepted for ICRA 2026
- Designed high fidelity **CAD models and prototypes of 12 iterations** of a cable driven surgical tool in **Solidworks**, characterized and validated final iterations in simulation and with physical prototypes
- Developed a high-fidelity **stereoscopic 3D simulation in Unity** with custom C# controllers to communicate via ROS
- Developed **Python/C++ control scripts for teleoperation** of simulated and real elbowed and wristed tools using DVRK MTMs through a custom inverse kinematic solver

PROJECTS

Tabletop Robotic Manipulator Platform

May 2026 – Present

- Developing a tabletop robotic manipulator platform as a generalized platform for manipulator control testing from scratch. Using tools such as **Yocto, RPi4, and custom designed cycloidal gearboxes** to achieve precise control via a **custom built RTOS**

Arm Team Software Lead, Simulations Lead, *Robotics For Space Exploration UofT*

Oct. 2022 – May 2026

Toronto, ON,

- Led the development of **end-to-end controls software in C++ and Python** for a 6 degree of freedom arm using **ROS2 and CANBUS protocols**, including joint space controls, inverse kinematic controls, and path planning algorithms
- Led the development of a **computer vision based autonomous keyboard typing algorithm** using a 6 degree of freedom arm, depth cameras, and object detection models

PUBLICATIONS

C. Balasubramanian, S. Zhang, T. Li, P. Kang, A. Nazari, T. Looi and D. Podolsky, "[Development of a Miniaturized 6-DoF Surgical Instrument with a 4-mm Elbow and 3-mm Wrist for Transoral Robotic Surgery](#)," in Proc. ICRA 2026, Vienna, Austria.

T. Li, S. Zhang, T. Looi and D. Podolsky, "[A Framework of Real2Sim Teleoperation System for Evaluating Surgical Robotic Tool Design](#)," in Proc. ISMR 2025, Atlanta, Georgia